MIL–P–274A
11 JULY 1956
SUPERSEDING
JAN–P–274
30 OCTOBER 1945

MILITARY SPECIFICATION

PAINT, LIQUID-VESICANT-DETECTOR M5

This specification has been approved by the Department of Defense, and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 **Scope.** This specification covers three types of paint for use in the detection of certain liquid vesicant agents.

1.2 **Classification.** M5 detector paint shall be of the following types as specified (see 6.1):

Type I — Cadmium-yellow (cadmium lithopone) pigment.

Type II — Hansa yellow G pigment.

Type III — Chrome yellow pigment.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR–C–96 — Cans, Steel (Tinned-Plate, Terne-Plate, and Black Sheet); Friction-Covers.

Cancelled OCT 1955

TT–C–83 — Cadmium-Yellow (Cadmium Lithopone); Dry (Paint-Pigment).

TT–C–290 — Chrome-Yellow and Chrome-Orange; Dry (Paint-Pigments).

TT–C–595 — Colors; (for) Ready-Mixed Paints.

TT–D–651 — Drier; Paint, Liquid.

TT–L–426 — Lithopone; Dry (Paint-Pigment).

TT–M–90 — Magnesium-Silicate; Dry (Paint-Pigment).

TT–O–367 — Oil; Linseed, Heat — Polymerized (Bodied), for Paint, Varnish, and Enamel.

TT–P–141 — Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

TT–T–291 — Thinner; Paint, Volatile Mineral Spirits (Petroleum-Spirits).

UU–T–111 — Tape, Paper, Gummed (Sealing and Securing).

FSC 6665

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LLL-B-631 — Boxes; Fiber, Corrugated (for Domestic Shipment).

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MIL-A-101 — Adhesive, Water Resistant, for Sealing Fiberboard Boxes.

JAN-P-106 — Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

JAN-P-108 — Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN-P-127 — Packaging and Packing for Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water-Resistant.

JAN-G-276 — Gum Ester.

MIL-B-10329 — B-1.

MIL-H-10330 — Hansa Yellow G.

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials.** All materials used in the manufacture of the paint shall conform to the applicable specifications forming a part of this specification.

3.2 **Physical.** The paint shall not settle or cake in the container to the extent that it cannot be broken up with a paddle to a smooth uniform paint of good brushing consistency. The paint shall conform to the physical requirements of table I when tested as specified therein.

TABLE I. *Physical requirements*

Property	Requirement	Test paragraph
Color	Olive Drab (3412 to 3433)	4.4.1
Weight	Minimum (pounds per gallon) Type I–13.0, Type II–9.5, Type III–12.0.	4.4.2
Adhesion	No cracking, chipping, or peeling off.	4.4.3
Coarse particles, skins and residue	Not more than 1.0 percent	4.4.4
Drying time and finish	Set to touch in not more than 1 hour. Dry hard in not more than 5 hours	4.4.5
Resistance to ultraviolet light	Panel shall not materially darken or fade and shall retain its sensitivity.	4.4.6
Resistance to heat	Panel shall not materially darken or fade and shall retain its sensitivity.	4.4.7

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3.3 Chemical.

3.3.1 *Composition.* The paint shall be prepared using ingredients in the proportions specified in table II. Each ingredient shall conform to the applicable paragraph of section 3.3.1 specified in table II.

TABLE II. *Composition*

Ingredient	Percent by weight			
	Type I	Type II	Type III	Applicable paragraph of section 3.3.1
Cadmium-yellow (Cadmium lithopone)	57 ± 1	.....	.....	3.3.1.1
Hansa yellow G	.....	11 ± 1	.....	3.3.1.2
Lithopone	.....	33 ± 1	.....	3.3.1.2
Chrome yellow	.....	.....	28 ± 1	3.3.1.3
Magnesium silicate	.....	.....	19 ±1	3.3.1.3
Vehicle	32	44	40	3.3.1.4
Mineral spirits	9	10	11	3.3.1.5
B–1	2	2	2	3.3.1.6
Drier	1 ounce per gallon of paint			3.3.1.7

3.3.1.1 *Pigment, type I paint.* The cadmium yellow (cadmium lithopone) shall conform to type II of Specification TT–C–83. The dry residues, on extraction, shall contain a minimum of 25.0 percent cadmium sulfide (CdS) when tested as specified in 4.4.8. There shall be no lead when tested as specified in 4.4.10.

3.3.1.2 *Pigment, type II paint.* The hansa yellow G shall conform to the requirements of Specification MIL–H–10330. The lithopone shall conform to the requirements of Specification TT–L–426. There shall be no lead when tested as specified in 4.4.10.

3.3.1.3 *Pigment, type III paint.* The chrome yellow pigment shall conform to the requirements for Type I of Specification TT–C–290. The magnesium silicate shall conform to the requirements of Specification TT–M–90, except that oil absorption shall be 60 and the particle size from 2 to 10 microns. The dry residue, on extraction, when tested as specified in 4.4.9, shall contain a minimum of 29.7 percent lead chromate (Pb Cr O4).

3.3.1.4 *Vehicle.* The vehicle shall be an ester gum-kettle bodied linseed oil type, of 10 gallon oil length, reduced with low solvency mineral spirits. The vehicle shall be free from lead, coal tar distillates, turpentine or pine oil and shall have a nonvolatile content of 48 to 52 percent, viscosity (Gardner and Holdt), A or B; color, 8 to 9; and acid number, 4 to 6 (see 6.2).

3.3.1.5 *Mineral spirits.* The mineral spirits shall be a low solvency petroleum product conforming to the requirements of Specification TT–T–291, and, in addition, shall show a kauri-butanol solvency of not over 40 ml and shall contain no coal tar distillates, pine oil or turpentine.

3.3.1.6 *Ingredient B–1.* The ingredient B–1 shall conform to Specification MIL–B–10329 (see 6.3).

3.3.1.7 *Drier.* The liquid drier shall be a manganese naphthenate drier conforming to the requirements of Specification TT–D–651 for type II, lead free drier, except that the

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drier shall contain 2.4 to 2.8 percent manganese when determined as metallic manganese, and shall contain no coal tar distillates, pine oil, or turpentine.

3.3.2 *Water content.* The paint shall contain not more than 0.5 percent water when tested as specified in 4.4.11.

3.3.3 *Sensitivity.* The paint shall meet the requirements for sensitivity when tested as specified in 4.4.12.

3.3.4 *Turpentine.* The paint shall contain no turpentine when tested as specified in 4.4.13.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of the material produced by one manufacturer in not more than 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials, providing the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.7).

4.2 Sampling. Eight four-ounce containers or one 32-ounce container shall be selected at random from each lot for test purposes. The contents of the 4-ounce containers shall be made up into one sample representing the lot. Each sample shall be tested separately as specified in 4.4.

4.3 Inspection. The inspector shall inspect the packaging, packing, and marking of the material for compliance with section 5.

4.4 Tests. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be conducted in parallel with each test, using the same quantities of reagents, and corrections shall be applied when significant.

4.4.1 *Color.* The paint shall be applied to a suitable nonabsorbent surface (glass plate or metal panel) with a coverage of approximately one square yard for 2 ounces, and allowed to dry for 24 hours. The color, as seen by diffused daylight, shall lie between 3412 and 3433 of Specification TT–C–595.

4.4.2 *Weight per gallon.* The weight per gallon of the paint shall be determined as specified in method 401.1 of Specification TT–P–141.

4.4.3 *Adhesion.* The paint (with a coverage of approximately 1 square yard for 2 ounces) shall be applied to thin, oil-free, vegetable parchment paper and allowed to dry for 24 hours. The painted paper shall then be folded (painted surface outwards), and a firm pressure supplied by thumb and forefinger to the folded edge.

4.4.4 *Coarse particles, skins, and residue.* The coarse particles, skins and residue shall be determined in accordance with method 409.1 prescribed in Specification TT–P–141.

4.4.5 *Drying time and finish.* The paint, after complete removal of any surface skin, shall be well mixed by shaking, stirring or both as may be necessary. The mixed paint shall then be brushed out on a clean, nonabsorbent surface (glass plate or metal panel) approximately 6 inches by 6 inches and exposed in a vertical position in a well ventilated room at 65° to 75°F. The painted surface shall be illuminated by diffused daylight for at least 4 hours during the drying period. In order to avoid interference through excessive humidity, care shall be taken that through the drying process, the temperature of the room is above the dew point. The paint shall have a satisfactory smooth, flat surface and shall dry within the time specified in table I.

4.4.6 *Resistance to ultra-violet light.* The paint (with a coverage of approximately 1 square yard for 2 ounces) shall be applied to thin, oil-free, vegetable parchment paper or solid index bristol stock, approximately 6 inches by 6 inches and allowed to dry at room temperature for 24 hours. The panel

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shall be exposed for 4 hours, at a distance of 15 inches, to the direct rays of 1 or more ultra-violet lamps. The total wattage used shall be a minimum of 720 and shall produce a minimum of 900 microwatts per square centimeter in a range from 2400 to 4000 Angstroms at a distance of 1 meter (see par. 6.6). After the 4 hours exposure, the panel shall not be materially darkened or faded when compared with a similar panel which has not been exposed to the ultra-violet light. The exposed panel shall show the required sensitivity when tested by the method described in paragraph 4.4.12.

4.4.7 *Resistance to heat.* The paint (with a coverage of approximately 1 square yard for 2 ounces) shall be applied to thin, oil-free, vegetable parchment paper or solid index bristol stock, approximately 6 inches by 6 inches, and dried for at least 8 hours in a well ventilated room. The painted panel shall be divided into 2 equal portions, 1 portion kept at room temperature and the other portion hung in an oven at a temperature of 60° C. (140° F.) for 5 hours. The panel shall be removed from the oven, cooled to room temperature and the sensitivity tested by the method described in paragraph 4.4.12. The sensitivity of the heat-treated panel shall be equal to that of the untreated panel. The color of the heat-treated panel shall not be appreciably darker or lighter than the color of a panel which has not been given the heat-treatment.

4.4.8 *Cadmium sulfide (type I paint).* The cadmium yellow shall be determined in accordance with the method specified in Specification TT–C–83.

4.4.9 *Lead chromate (type III paint).* Lead chromate shall be determined in accordance with the method specified in Specification TT–C–290.

4.4.10 *Lead.* The lead content shall be determined in accordance with the method specified in Specification TT–D–651.

4.4.11 *Water content.* The water content of the paint shall be determined as specified in method 408.1 of Specification TT–P–141.

4.4.12 *Sensitivity.* The paint (with a coverage of approximately 1 square yard for 2 ounces) shall be applied to thin, oil-free, vegetable parchment paper, or other non-absorbent surface, and allowed to dry for 24 hours at existing temperature. A vivid crimson color shall be produced within 5 seconds when a drop, no greater than 2 mm. (0.0787 inch) diameter, of methyl salicylate is allowed to fall on the surface of the paint film. This crimson color shall remain conspicuous for at least 30 minutes.

4.4.13 *Turpentine.* Approximately 150 gm of the paint shall be transferred to a 500 ml. flask fitted with a 2-hole cork stopper carrying through 1 hole a spray trap connected with a vertical condenser. An influx tube for steam shall be passed through the other hole in the stopper. (This tube should dip below the surface of the paint.). The flask shall be heated in an oil bath or an air bath at 100° to 105° C. and a current of steam passed through the flask. With the steam still passing through, the temperature of the bath shall be raised to 130° to 135° C. The distillate shall be caught in a small separatory funnel and distillation continued until 300 ml. of water have been condensed. Portions of this water may be drawn from the cork of the separatory funnel from time to time, but care must be taken not to draw out any of the volatile thinner. The distillate shall be allowed to stand until it separates into 2 layers, then the water drawn off and the volatile thinner filtered through a dry filter paper into a dry flask. Two drops of the distillate and 2 to 3 ml. of chloroform shall be placed in a dry test tube and 1 drop of antimony pentachloride added. A slow or slight change in color will indicate the absence of turpentine. A rapid change in color to a dark red or purple will indicate the presence of turpentine.

4.5 **Rejection and resubmission.** If any specimen fails to conform to this specifica-

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tion, the lot represented shall be rejected. The contractor, at no expense to the Government, shall have the option of having a partial or complete analysis made of each container in the rejected lot. He may then remove the nonconforming containers and resubmit the remaining portion of the lot for acceptance testing. If any specimen from the resubmitted lot fails to conform to this specification, the lot shall be finally rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 *Level A.*

5.1.1.1 *Four-ounce quantity.* The material shall be packaged in a 4-ounce container conforming to Specification RR–C–96. The containers shall be packaged in 2 layers, 6 rows, and 8 containers per row in a W5c, style RSC, fiberboard box conforming to Specification JAN–P–108. Interlocking partitions forming snug fitting cells shall be used to separate the containers along with a full intermediate pad made of corrugated fiberboard conforming to compliance symbol W6c of Specification Jan–P–108. The flaps of the fiberboard carton shall be sealed with water-resistant adhesive conforming to Specification MIL–A–101. All seams of the carton shall be closed with three inch wide, grade B, type II, water-resistant, pressure-sensitive, adhesive tape, conforming to Specification JAN–P–127.

5.1.1.2 *Thirty-two-ounce quantity.* The material shall be packaged in a 32-ounce container conforming to Specification RR–C–96. The containers shall be packaged in 1 layer, 3 rows, and 4 containers per row in a W5c, style RSC, fiberboard box conforming to Specification JAN–P–108. Interlocking partitions forming snug fitting cells shall be used to separate the containers made of corrugated fiberboard conforming to compliance symbol W6c of Specification JAN–P–108. The flaps of the fiberboard carton shall be sealed with water-resistant adhesive conforming to Specification MIL–A–101. All seams of the carton shall be closed with three-inch wide, grade B, type II, water-resistant, pressure-sensitive, adhesive tape, conforming to Specification JAN–P–127.

5.1.2 *Level C.* Same as 5.1.1 except that the fiberboard box shall conform to Specification LLL–B–631 and the tape shall conform to Specification UU–T–111.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Wood box.* The 4-ounce containers packaged in accordance with 5.1.1.1 or the 32-ounce containers packaged in accordance with 5.1.1.2 shall be packed in a style 4, nailed wood box conforming to the applicable requirements of Specification JAN–P–106 and closed and strapped in accordance with the appendix, thereto. The gross weight of the wood box and contents shall not exceed seventy pounds.

5.2.1.2 *Fiberboard box.* The 4-ounce containers packaged in accordance with 5.1.1.1 or the 32-ounce containers packaged in accordance with 5.1.1.2 shall be packed in a V3s, style RSC, fiberboard box conforming to Specification JAN–P–108. The fiberboard box shall be sealed in accordance with 5.1.1.1 and strapped in accordance with Specification JAN–P–108.

5.2.2 *Level B.* The packing shall be the same as the packaging requirements specified in paragraph 5.1.1.

5.2.3 *Level C.* The packing shall be the same as the packaging requirements specified in paragraph 5.1.2.

5.3 **Marking.** In addition to any special marking required by the contract or order, all cans and interior and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129. In addition marking shall include instructions

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as follows: Apply paint by brushing or spraying on any surface suitable for painting such as metal, wood or paper. Use only pure mineral spirits for thinning. Outdoor painting should be carried out in the evening to permit overnight drying. Under ning to permit overnight drying. Under severe conditions of exposure, surfaces should be repainted every 2 or 3 weeks.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify:

a. Title, number, and date of this specification.

b. Type of paint desired (see 1.2).

c. Unit package size desired.

d. Level of packaging and packing required.

6.2 **Vehicle.** A varnish made according to the following procedure has been found to be satisfactory for this purpose:

Ester gum (Grade I, Class a) of:

Specification JAN–G–276	100 pounds
Kettle-bodied linseed oil	10 gallons
Mineral spirits	28 gallons

Heat the ester gum and oil to 575° F. and hold at this temperature for body, approximately 90 to 120 minutes. Cool and reduce with mineral spirits.

6.3 **Method of mixing ingredient B–1.** The cadmium lithopone (Type I pigment) or the hansa yellow G and lithopone (Type II pigment) should be thoroughly ground on a water-cooled roller mill with the bulk of the vehicle, before the addition of the ingredient B–1. The chrome yellow (Type III pigment) should be thoroughly ground on a water-cooled roller mill with the bulk of the vehicle before the addition of the ingredient B–1. The magnesium silicate should be added with stirring only, grinding in not necessary. The ingredient B–1 should be made into a paste with the remainder of the vehicle, using a water-cooled roller mill at light pressure (attention is directed to the fact that excess heat and excess grinding of ingredient B–1 will adversely affect the color of the finished paint) and the paste then thoroughly incorporated with the pigment-vehicle portion and the mineral spirits in a mixer equipped with a stirrer. The ingredients should be at room temperature when incorporated.

6.4 The suggested method of manufacture in paragraphs 6.2 and 6.3 are furnished to the manufacturer, with the distinct understanding (a) that the manufacturer is not required to follow the methods unless so stated in the contract or order; (b) that the provisions of this specification must be met irrespective of the method of manufacture; and (c) that the Government makes no warranty that material made in accordance with the methods will pass the specification requirements.

6.5 A generally satisfactory paint of spraying consistency may be prepared by adding approximately 4 parts by weight of mineral spirits to each 100 parts by weight of the paint.

6.6. Ultra-violet light of the characteristics specified in paragraph 4.4.6 may be obtained by employing two UA–2GA2 tubular A. C. Uviarc outfits furnished by the Nela Specialty Division, General Electric Company, Hoboken, New Jersey.

6.7 **Batch.** A batch is defined as that quality of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.8 **Sampling and testing.** When the contractor consistently produces high quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However,

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the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

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Custodian:

Army—Chemical Corps
Navy—Bureau of Yards and Docks
Air Force

Other interests:

Army—M
Navy—MC, Md

Preparing activity:

Army—Chemical Corps

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